

Stage 3A Physics Note: Initial Data for the 5D Uniform Black String

GL-with-AI Project

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1 Scope and Status

This note is a polished physics review artifact for Stage 3A. It records the target continuum initial data and design choices for a future implementation of the five-dimensional uniform black string and a Gregory-Laflamme seed perturbation.

This is design only. The initial data described here are not implemented in `BlackStringToy`. The current `BlackStringToy` scaffold remains a non-physical build/run scaffold derived from the public GRChombo `Examples/BinaryBH` example. Nothing in this note claims that the current code evolves a five-dimensional black string.

2 Target Geometry

The target unperturbed spacetime is the five-dimensional uniform black string, formally

$$\text{Schwarzschild}_4 \times S_z^1. \tag{1}$$

The dimensional bookkeeping for the intended reduced evolution is:

- physical spacetime dimension: 5;
- physical spatial dimension: `GR.SPACEDIM` = 4;
- Chombo gridded dimension: `CH.SPACEDIM` = 2;
- gridded directions: a radial/cartoon coordinate x and compact string coordinate z ;
- hidden/reconstructed directions: the transverse S^2 , represented by $\text{SO}(3)$ /modified-cartoon variables and source terms.

3 GP-like Horizon-Penetrating Form

The planned continuum starting point is a horizon-penetrating Gullstrand-Painleve-like form of the uniform string. A schematic line element is

$$ds^2 = -dt^2 + \left(dx + s \sqrt{\frac{r_0}{x}} dt \right)^2 + x^2 d\Omega_2^2 + dz^2. \tag{2}$$

Here $s = \pm 1$ labels the GP branch/sign convention. The slice is horizon-penetrating: the horizon at $x = r_0$ is not a coordinate singularity on the spatial slice.

The sign s , the time orientation, and GRChombo's convention for extrinsic curvature K_{ij} must be matched before implementation. All extrinsic-curvature formulas below should be read with that caveat.

4 ADM Variables and GP Extrinsic Curvature

The intrinsic spatial metric on the GP-like slice is flat in the radial transverse sector times the compact string direction:

$$dl^2 = dx^2 + x^2 d\Omega_2^2 + dz^2. \quad (3)$$

The natural GP lapse and shift associated with Eq. (2) are

$$\alpha = 1, \quad \beta^x = s\sqrt{\frac{r_0}{x}}, \quad \beta^z = 0. \quad (4)$$

Using the convention

$$K_{ij} = \frac{1}{2\alpha} (D_i \beta_j + D_j \beta_i - \partial_t \gamma_{ij}), \quad (5)$$

and the time-independent spatial metric, the nonzero components for the uniform string are, up to GRChombo's possible overall K_{ij} sign convention,

$$K_{xx} = -\frac{s}{2} \sqrt{\frac{r_0}{x^3}}, \quad (6)$$

$$K_{ww} = s\sqrt{\frac{r_0}{x^3}} \quad \text{for each hidden transverse direction,} \quad (7)$$

$$K_{zz} = 0. \quad (8)$$

Thus $K_{zz} = 0$, but the full four-spatial-dimensional trace is not zero. The trace is

$$K = K^x_x + K^z_z + K^{w_1}_{w_1} + K^{w_2}_{w_2} = \frac{3s}{2} \sqrt{\frac{r_0}{x^3}}, \quad (9)$$

again up to the overall GRChombo K_{ij} sign convention. If GRChombo uses the opposite convention to Eq. (5), then all K_{ij} , K , and conformal traceless components A_{ij} and A_{ww} must be sign-flipped.

5 Scalar and Sign Note for K

The quantity $K = \gamma^{ij} K_{ij}$ is a scalar on a fixed spatial slice, so it is invariant under spatial coordinate changes on that slice. Its sign can still change if one reverses the GP branch, reverses the time orientation, or uses the opposite convention for the future-pointing normal. The two values $s = \pm 1$ are time-reversed ingoing/outgoing embeddings of the same flat intrinsic spatial geometry. Therefore the sign of K is fixed only after choosing both the GP branch and GRChombo's K_{ij} convention.

6 Conformal CCZ4 Variable Map

The target variables follow the conformal CCZ4 convention already recorded in the implementation notes:

$$\gamma_{ij} = \chi^{-1} h_{ij}, \quad \gamma_{ww} = \chi^{-1} h_{ww}. \quad (10)$$

The hidden physical extrinsic-curvature component is reconstructed as

$$K_{ww} = \chi^{-1} \left(A_{ww} + \frac{h_{ww} K}{\text{GR_SPACEDIM}} \right). \quad (11)$$

For $\text{GR_SPACEDIM} = 4$,

$$A_{ij} = \chi \left(K_{ij} - \frac{1}{4} \gamma_{ij} K \right), \quad A_{ww} = \chi K_{ww} - \frac{1}{4} h_{ww} K. \quad (12)$$

At design level, the future initial-data class must provide:

- χ , initially 1 before the GL perturbation;
- conformal metric components h_{xx} , h_{zz} , and h_{xz} ;
- hidden conformal metric component h_{ww} ;
- full trace K , using all four spatial directions;
- traceless components A_{xx} , A_{zz} , A_{xz} , and A_{ww} ;
- lapse, shift, Gamma variables, and gauge-driver variables consistent with the chosen startup convention.

For the unperturbed flat GP-like spatial slice, the expected conformal metric starting values are

$$h_{xx} = 1, \quad h_{zz} = 1, \quad h_{xz} = 0, \quad h_{ww} = 1, \quad (13)$$

before any GL perturbation is applied.

7 Meaning of $h_{ww} = 1$

The variable h_{ww} is not the angular metric component $g_{\theta\theta}$. It is the hidden Cartesian-like conformal metric component used by the $\text{SO}(3)$ /modified-cartoon reduction. In flat transverse space, the suppressed Cartesian directions have $\gamma_{ww} = 1$.

The angular factor x^2 in $x^2 d\Omega_2^2$ is supplied by the coordinate/cartoon reconstruction. The physical transverse sphere radius is

$$R = x \sqrt{\gamma_{ww}} = x \sqrt{\chi^{-1} h_{ww}}. \quad (14)$$

Setting $h_{ww} = x^2$ would double-count the angular radius.

8 Horizon Representation

The future apparent-horizon representation is

$$x = h(t, z). \quad (15)$$

Here $h(t, z)$ is the coordinate radius/location of the apparent horizon on the reduced grid. For the unperturbed uniform string,

$$h(0, z) = r_0. \quad (16)$$

During Gregory-Laflamme evolution, $h(t, z)$ becomes z -dependent. The physical transverse S^2 radius on the horizon is

$$R(t, z) = h(t, z) \sqrt{\gamma_{ww}(t, h(t, z), z)}. \quad (17)$$

9 GL Seed Perturbation

The planned Gregory-Laflamme seed from the reproduction roadmap is a conformal-factor perturbation localized near the horizon:

$$\chi = 1 + \epsilon \sin\left(\frac{2\pi n z}{L}\right) \exp\left[-\left(\frac{x}{r_0} - \frac{r_0}{x}\right)^2\right]. \quad (18)$$

Planned defaults are

$$\epsilon = 0.01, \quad n = 1, \quad r_0 = 1, \quad L/r_0 = 10. \quad (19)$$

This perturbation is planned but not implemented. It intentionally introduces small constraint violations that must be monitored and damped before nonlinear GL growth dominates.

10 Natural GP Shift Versus Zero-Shift Gauge Startup

The GP line element is used primarily to define a regular horizon-penetrating initial hypersurface and to derive the geometric data γ_{ij} and K_{ij} . Natural GP coordinates have the nonzero shift in Eq. (4).

The evolution gauge variables do not necessarily need to be initialized to the natural GP shift. If following the reproduction roadmap, one can initialize

$$\alpha(0) = 1, \quad \beta^i(0) = 0, \quad (20)$$

while keeping γ_{ij} and nonzero K_{ij} from the GP slice. In that case the data are GP-slice geometric data, not a fully stationary GP-coordinate evolution.

Do not set $K_{ij} = 0$ merely because the evolution shift is initialized to zero. K_{ij} is part of the geometric initial data and must not be recomputed from the chosen gauge-startup shift.

11 Turduckening and Interior Handling

The GP-like expressions include powers such as $\sqrt{r_0/x}$, so a cutoff or smoothing strategy is needed near $x = 0$. The planned cutoff scale is

$$x_{\text{cut}} = 0.1r_0. \quad (21)$$

The quantities that require an interior prescription include χ , the GL perturbation, the shift if used geometrically, K , A_{ij} , A_{ww} , h_{ww} , and derived Gamma/constraint quantities. The exact smoothing profile is unresolved. A hard clamp is simple but may create derivative artifacts, so a smoother transition may be needed.

12 Boundary and Periodicity

The compact string direction should be periodic:

$$z \sim z + L. \quad (22)$$

The outer radial boundary should be far enough away that it is not causally connected to the string during the target evolution window. The reproduction roadmap uses $x_{\text{outer}} = 256r_0$. The long-term outer boundary will need a reviewed outgoing or Sommerfeld-like treatment.

The cartoon axis and radial-origin regularity are separate issues. The inherited BinaryBH reflective/Sommerfeld choices, binary-puncture tracking, puncture-centered tagging, and Weyl4 extraction settings are not validated long-term assumptions for the reduced 5D black-string system.

13 Unresolved Decisions Before Coding

- Choose the GP branch s and match it to GRChombo's K_{ij} sign convention.
- Confirm whether to follow the roadmap startup $\alpha(0) = 1$, $\beta^i(0) = 0$ while using GP-slice γ_{ij} and nonzero K_{ij} .
- Decide and verify the exact **hww/Aww** enum layout, including the AH convention `c_hww = c_K - 1`.
- Establish a reliable `GR_SPACEDIM = 4`, `CH_SPACEDIM = 2` build path.
- Define unperturbed uniform-string Hamiltonian and momentum constraint validation tests.
- Compare horizon radius and area against r_0 and $A_0 = 4\pi r_0^2 L$.
- Choose and validate the turduckening smoothing profile.